

Table 1: Performance of various algorithms for class/feature unlearning with VAEs on MNIST and CelebA and diffusion transformers on ImageNet-1K. Each experiment is repeated 10 times, and the standard deviations are shown in parentheses. FID values are calculated using 25,000 samples for MNIST and CelebA and 22,000 samples for ImageNet-1K. **✗** indicates that the generated samples after unlearning are unrecognizably different from the original model. Bold indicates the best score among instances with acceptable FID. **✓** indicates the generated samples after unlearning are perceptually indistinguishable from the original model in terms of visual fidelity. An asterisk (*) denotes cases where the algorithm failed to reach the target fraction of forget samples in the generated images within the allotted training steps. We only report on (A) and (A-D) for MNIST since they demonstrate catastrophic forgetting in all cases; for CelebA (A) and (A-D) result in NaN for FID, generating white images. The original models for MNIST, CelebA and ImageNet-1K have FID 20.7, 166.3 and 12.7 respectively, and we are unlearning the digit 1, faces identified as “male” and images of class 207 (golden retriever) respectively.

Dataset	Algorithm	Time to unlearn (s) ↓	FID ↓	Δt (ms)	5-step f-ratio ↓
MNIST	Gradient Ascent (A)	0.024 (0.004)	612.3 (4.9) ✗	5.1 (0.47)	0.00 (0.01)
	Ascent descent (A-D)	0.025 (0.006)	266.9 (19.3) ✗	5.1 (0.09)	0.01 (0.01)
	Gradient surgery (S)	1.016 (0.907)	23.0 (0.2) ✓	7.1 (0.06)	0.05 (0.03)
	UNO	0.055 (0.009)	21.8 (0.2) ✓	20.5 (0.27)	0.00 (0.00)
	UNO-S	0.041 (0.010)	21.8 (0.4) ✓	15.1 (0.35)	0.00 (0.00)
	Moon et al. (2024)	0.065 (0.052)	25.1 (0.4) ✓	5.3 (0.04)	0.03 (0.02)
CelebA	Gradient surgery (S)	10.750* (3.362)	176.0 (3.7) ✓	18.3 (0.18)	0.11 (0.05)
	UNO	0.527 (0.003)	174.3 (1.6) ✓	175.7 (0.91)	0.00 (0.00)
	UNO-S	0.283 (0.040)	177.1 (4.7) ✓	101.0 (0.35)	0.00 (0.00)
	Moon et al. (2024)	7.827 (3.683)	172.9 (0.8) ✓	15.4 (0.04)	0.14 (0.03)
ImageNet-1K	Gradient surgery (S)	7.622 (2.211)	12.3 (0.8) ✓	712.4 (8.99)	0.70 (0.28)
	UNO	6.361 (0.909)	11.8 (0.8) ✓	1817.3 (4.59)	0.00 (0.00)
	UNO-S	6.495 (1.928)	12.0 (0.8) ✓	1273.4 (8.04)	0.07 (0.12)
	ESD-u-1 Gandikota et al. (2023)	6.600 (1.299)	11.6 (0.6) ✓	362.6 (4.96)	0.63 (0.20)
	ESD-u-3 Gandikota et al. (2023)	2.023 (0.677)	13.1 (1.0) ✓	348.8 (5.64)	0.14 (0.10)
	SalUn	4.411 (0.766)	242.2 (3.2) ✓	711.4 (3.36)	0.13 (0.09)